

List of Formulas

① Standard eqⁿ of FM wave

$$V_{fm}(t) = V_c \cos[\omega_c t + m_f \sin \omega_m t]$$

② $\Delta f = K_f V_m \dots (\text{Hz})$

③ $m_f = \frac{\Delta f}{f_m} \rightarrow \text{Modulation index}$

④ $K_f = \frac{\Delta \omega}{\Delta v} \left(\frac{\text{rad/sec}}{v} \right)$

⑤ $K_p = \frac{\Delta \theta}{\Delta v} \left(\frac{\text{rad/v}}{v} \right)$

⑥ % Modulation = $\frac{\Delta f(\text{actual})}{\Delta f(\text{max})} \times 100$

⑦ $B = 2 f_m \rightarrow m_f \leq 1$

⑧ $B = 2 \Delta f \rightarrow m_f > 1$

⑨ $B = 2(n \times f_m)$

⑩ $B = 2(\Delta f + f_m) \rightarrow \text{Carson's rule}$

Bandwidth

⑪ Deviation ratio = $\frac{\Delta f(\max)}{f_m(\max)}$

⑫ Carrier Swing
(peak to peak deviation) = $f_c + \Delta f - (f_c - \Delta f)$
= $\boxed{2\Delta f}$

⑬ $P_c = \frac{V_c^2}{2R}$

⑭ Power = $\frac{V_{rms}^2}{R} = \frac{(V_p/\sqrt{2})^2}{R}$

Example 1

A sinusoidal modulating waveform of amplitude 5V and a frequency of 2 KHz is applied to FM generator, which has a frequency sensitivity of 40Hz/volt. Calculate the frequency deviation, modulation index, and bandwidth.

$$V_m = 5V, f_m = 2 \text{ KHz}$$

$$K_f = 40 \text{ Hz/volt}$$

$$\Delta f = ? \quad m_f = ? \quad \text{B.W} = ?$$

$$\Delta f = K_f V_m = 40 \times 5 = \boxed{200 \text{ Hz}}$$

$$m_f = \frac{\Delta f}{f_m} = \frac{200}{2 \times 10^3}$$

$$\boxed{m_f = 0.1} \rightarrow \underline{m_f < 1}$$

$$\begin{aligned} \text{B.W} &= 2f_m \\ &= 2 \times 2 \times 10^3 \end{aligned}$$

$$\boxed{\text{B.W} = 4 \text{ KHz}}$$

Example 2

An FM wave is given by

$$s(t) = \underline{20} \cos(\overbrace{8\pi \times 10^6 t}^{\omega_c t} + \underline{9} \sin(\overbrace{2\pi \times 10^3 t}^{\omega_m t}))$$

Calculate the frequency deviation, bandwidth and power of FM wave

$\Delta f = ?$ $B.W = ?$ $Power = ?$

Compare $V_{fm}(t) = \underline{V_c} \cos(\omega_c t + m_f \sin \omega_m t)$

$V_c = 20V$

$m_f = 9$

$m_f = \frac{\Delta f}{f_m}$

$2\pi f_m = 2\pi \times 10^3$

$f_m = 1 \times 10^3$

$f_m = 1 \text{ KHz}$

$2\pi f_c t = 8\pi \times 10^6 t$

$f_c = \frac{8\pi \times 10^6 t}{2\pi t} \therefore f_c = 4 \text{ MHz}$

Example 2

An FM wave is given by

$$s(t) = 20 \cos(8\pi \times 10^6 t + 9 \sin(2\pi \times 10^3 t))$$

Calculate the frequency deviation, bandwidth and power of FM wave

$$m_f = \frac{\Delta f}{f_m}$$

$$\Delta f = m_f f_m$$

$$\Delta f = 9 \times 1 \times 10^3$$

$$\Delta f = 9 \text{ KHz}$$

$$m_f = 9 \rightarrow m_f > 1$$

$$B.W = 2(\Delta f + f_m)$$

$$B.W = 2(9 \times 10^3 + 1 \times 10^3)$$

$$B.W = 20 \text{ KHz}$$

$$B.W = 2\Delta f$$

$$B.W = 2 \times 9 \times 10^3$$

$$B.W = 18 \text{ KHz}$$

Any of the formula can be used

Example 2

An FM wave is given by $\rightarrow V_c$

$$s(t) = \underline{\underline{20}} \cos(8\pi \times 10^6 t + 9 \sin(2\pi \times 10^3 t))$$

Calculate the frequency deviation, bandwidth and power of FM wave

$$\text{Power} = \frac{V_c^2}{2R}$$

$$\text{Assuming } R = 1 \Omega$$

$$\text{Power} = \frac{(20)^2}{2(1)} = \frac{400}{2}$$

$$\boxed{\text{Power} = 200 \text{ Watts}}$$

Example 3

An FM wave is given by

$$v(t) = 12 \cos(6 \times 10^8 t + 5 \sin 1250 t)$$

Determine (i) carrier frequency (ii) modulation index (iii) Modulation frequency (iv) Maximum deviation (v) power dissipated in 10 ohm R.

$$f_c = 95.49 \text{ MHz}$$

$$m_f = 5$$

$$F_m = 199 \text{ Hz}$$

$$\Delta f = 995 \text{ Hz}$$

$$\text{Power} = 7.2 \text{ W}$$

Example 4

Find the instantaneous frequency in Hz for the given signal:

$$10\cos\left(200\pi t + \frac{\pi}{3}\right)$$

Comparing with

$$f_m(t) = V_c \cos \theta_i(t)$$

$$\omega_i = \frac{d\theta_i(t)}{dt}$$

$$\theta_i(t) = 200\pi t + \frac{\pi}{3}$$

$$\omega_i = \frac{d}{dt} \left[200\pi t + \frac{\pi}{3} \right]$$

$$\omega_i = 200\pi \text{ rad/sec}$$

$\rightarrow \theta_i$

$$f_i = \frac{\omega_i}{2\pi}$$

$$f_i = \frac{200\pi}{2\pi}$$

$$f_i = 100 \text{ Hz}$$

Example 5

$$\omega_i = \frac{d\phi_i}{dt}$$

Find the instantaneous frequency in Hz for the given signal:

$$10\cos(20\pi t + \pi t^2)$$

$$\phi_i(t) = 20\pi t + \pi t^2$$

$$\omega_i = \frac{d}{dt} [20\pi t + \pi t^2]$$

$$\omega_i = 20\pi + 2\pi t$$

$$\omega_i = \pi [20 + 2t]$$

$$f_i = \frac{\omega_i}{2\pi}$$

$$f_i = \frac{\pi [20 + 2t]}{2\pi}$$

$$f_i = 10 + t \text{ Hz}$$

$$\text{at } t = 0 \rightarrow f_i = 10 \text{ Hz}$$

$$t = 1 \text{ Sec} \rightarrow f_i = 11 \text{ Hz}$$

$$t = 2 \text{ Sec} \rightarrow f_i = 12 \text{ Hz}$$

Example 6

Find the instantaneous frequency in Hz for the given signal:

$$\underbrace{\cos 200\pi t}_A \cdot \underbrace{\cos(5\sin 2\pi t)}_B + \underbrace{\sin 200\pi t}_A \cdot \underbrace{\sin(5\sin 2\pi t)}_B$$

$$\cos(A-B)$$

$$A = 200\pi t$$

$$B = 5\sin 2\pi t$$

$$\theta_i(t) = (200\pi t - 5\sin 2\pi t)$$

$$\omega_i = \frac{d}{dt} [(200\pi t) - 5\sin(2\pi t)]$$

$$\omega_i = \pi (200 - 10 \cos(2\pi t))$$

$$f_i = 100 - 5 \cos(2\pi t) \text{ Hz}$$

Example 7

A 107.6 MHz carrier signal is frequency modulated by a 7 KHz sine wave. The resultant FM signal has a frequency deviation of 50 KHz. Determine: (i) carrier swing of FM signal (ii) highest and lowest frequencies attained by the modulated signal (iii) the modulation index of FM wave

$$F_c = 107.6 \text{ MHz} \quad f_m = 7 \text{ KHz}$$

$$\Delta f = 50 \text{ KHz}$$

$$\begin{aligned}
 \text{(i) Carrier swing} &= 2\Delta f \\
 &= 100 \text{ KHz}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) Highest freq} &= F_c + \Delta f \\
 &= 107.6 \text{ MHz} + 50 \text{ KHz}
 \end{aligned}$$

$$= 107.65 \text{ MHz}$$

$$\begin{aligned}
 \text{(iii) Lowest freq} &= F_c - \Delta f \\
 &= 107.6 \text{ MHz} - 50 \text{ KHz} \\
 &= 107.55 \text{ MHz}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) } m_f &= \frac{\Delta f}{f_m} = \frac{50 \times 10^3}{7 \times 10^3} \\
 m_f &= 7.14
 \end{aligned}$$

Example 8

What is the modulation index of an FM signal having a carrier of 100 KHz when the modulating signal has a frequency of 8 KHz?

$$\text{Carrier Swing} = 2\Delta f$$

$$100 \text{ KHz} = 2\Delta f$$

$$\Delta f = 50 \text{ KHz}$$

$$m_f = \frac{\Delta f}{f_m} = \frac{50 \times 10^3}{8 \times 10^3} = 6.25$$

Example 9

An FM transmission has a frequency deviation of 20 KHz. Determine:

- % modulation of this signal if it is broadcasted in 88-108 MHz band
- Calculate the % modulation if this signal is broadcasted as the audio portion of a TV broadcast

Assuming $\Delta f = 75 \text{ KHz}$

$$\Delta f = 20 \text{ KHz}$$

$$(i) \quad \% \text{ modulation} = \frac{(\Delta f)_{\text{actual}}}{(\Delta f)_{\text{maximum}}} = \frac{20 \times 10^3}{(75 \times 10^3)} = 26.67\%$$

$$(ii) \quad \% \text{ modulation} = \frac{20 \times 10^3}{(25 \times 10^3)} = 80\%$$

Assume $\leftarrow (25 \times 10^3)$

Example 10

Find the bandwidth of a commercial FM transmission if frequency deviation is 75 KHz and modulating frequency is 15 KHz (*Carson's Rule*)

$$\begin{aligned} B.W &= 2(\Delta f + f_m) \\ &= 2(75 \text{ KHz} + 15 \text{ KHz}) \end{aligned}$$

$$B.W = 180 \text{ KHz}$$

$$\begin{aligned} B.W &= 2\Delta f \\ B.W &= 2(75 \times 10^3) \end{aligned}$$

$$B.W = 150 \text{ KHz}$$

Example 11

Determine the bandwidth of a narrowband FM signal which is generated by a 4 KHz audio signal modulating a 125 MHz carrier.

$$\begin{aligned} B.W &= 2 f_m \\ &= 2 \times 4 \times 10^3 \\ \boxed{B.W &= 8 \text{ KHz}} \end{aligned}$$

Example 12

In a FM system, the modulating frequency is 1 KHz, the modulating voltage 2V and the deviation is 6KHz. If the modulating voltage is raised to 4V, then what will be the new deviation? If the modulating voltage is further raised to 8V and the modulating frequency is reduced to 500 Hz, what will be new deviation?

$$f_m = 1 \text{ KHz} \quad V_m = 2 \text{ V} \quad \Delta f = 6 \text{ KHz}$$

Case-I $V_m = 4 \text{ V} ; \Delta f = ?$

$$\Delta f = K_f V_m = 3 \times 4 = \boxed{12 \text{ KHz}} \quad \textcircled{1}$$

Case-II $V_m = 8 \text{ V} ; \Delta f = ?$

$$\Delta f = K_f V_m = 3 \times 8 = \boxed{24 \text{ KHz}} \quad \textcircled{2}$$

$$K_f = \frac{\Delta f}{V_m} = \frac{6 \text{ KHz}}{2}$$

$$\boxed{K_f = 3 \text{ KHz}}$$

$$m_f = \frac{\Delta f}{f_m} = \frac{24 \times 10^3}{500}$$

$$\boxed{m_f = 48}$$

Example 13

For the same data of the Example 12, calculate the modulation index in each case. Comment on the result.